We will now prove our first result — Schur's lemma. Although it is very easy to prove, it is fundamental in the whole subject of representation theory.

Proposition 2.3.9 (Schur's lemma)**.** *Let V_1, V_2 be representations of an algebra A over any field F (which need not be algebraically closed). Let $\phi : V_1 \to V_2$ be a nonzero homomorphism of representations. Then:*

(i) If V_1 is irreducible, ϕ is injective.

(ii) If V_2 is irreducible, ϕ is surjective.

Thus, if both V_1 and V_2 are irreducible, ϕ is an isomorphism.

Proof. (i) The kernel K of ϕ is a subrepresentation of V_1. Since $\phi \neq 0$, this subrepresentation cannot be V_1. So by irreducibility of V_1 we have $K = 0$.

(ii) The image I of ϕ is a subrepresentation of V_2. Since $\phi \neq 0$, this subrepresentation cannot be 0. So by irreducibility of V_2 we have $I = V_2$. □

Corollary 2.3.10 (Schur's lemma for algebraically closed fields)**.** *Let V be a finite dimensional irreducible representation of an algebra A over an algebraically closed field k, and let $\phi : V \to V$ be an intertwining operator. Then $\phi = \lambda \cdot \mathrm{Id}$ for some $\lambda \in k$ (a scalar operator).*

Remark 2.3.11. Note that this corollary is false over the field of real numbers: it suffices to take $A = \mathbb{C}$ (regarded as an $\mathbb{R}$-algebra) and $V = A$.

Proof. Let λ be an eigenvalue of ϕ (a root of the characteristic polynomial of ϕ). It exists since k is an algebraically closed field. Then the operator $\phi - \lambda\,\mathrm{Id}$ is an intertwining operator $V \to V$, which is not an isomorphism (since its determinant is zero). Thus by Proposition 2.3.9 this operator is zero, hence the result. □

Corollary 2.3.12. *Let A be a commutative algebra. Then every irreducible finite dimensional representation V of A is 1-dimensional.*

Remark 2.3.13. Note that a 1-dimensional representation of any algebra is automatically irreducible.